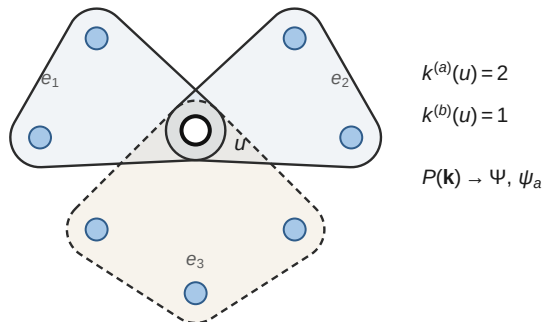
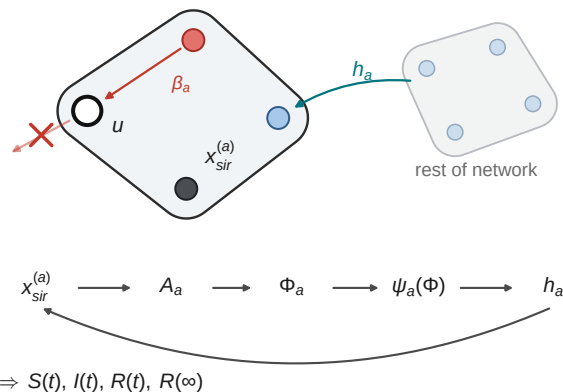


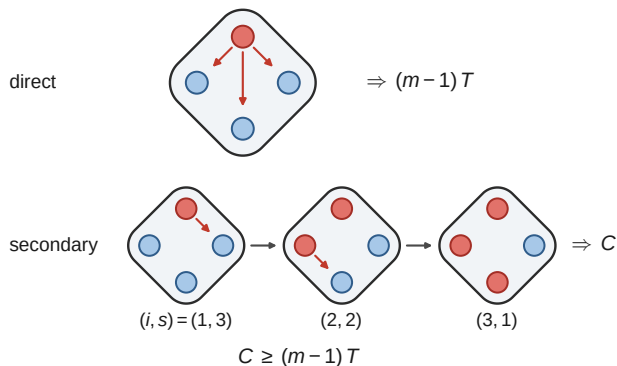
(a) Multiplex hypergraph



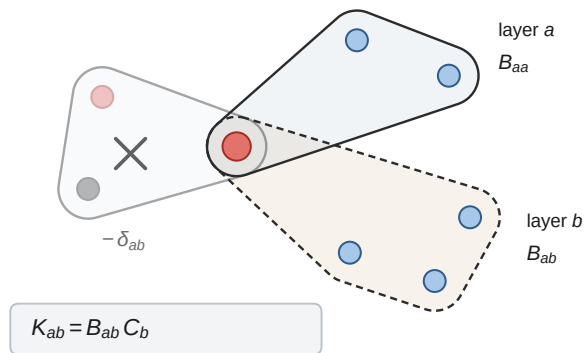
(b) Edge-based closure



(c) Within-hyperedge factor C



(d) Excess hyperdegree B



● susceptible ● infected ● recovered test node u

layer a layer b